PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 10 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

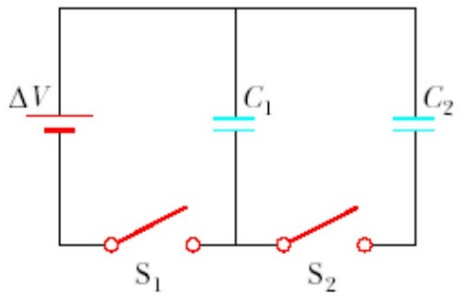
Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Tutorial 6 - Question 23

23. Consider the circuit shown in Figure P26.23, where $C_1 = 6.00\ \mu F$, $C_2 = 3.00\ \mu F$, and $\Delta V = 20.0$ V. Capacitor C_1 is first charged by the closing of switch S_1. Switch S_1 is then opened, and the charged capacitor is connected to the uncharged capacitor by the closing of S_2. Calculate the initial charge acquired by C_1 and the final charge on each capacitor.



2. [____] Tutorial 6 - Question 67

67. An isolated capacitor of unknown capacitance has been charged to a potential difference of 100 V. When the charged capacitor is then connected in parallel to an uncharged 10.0-μF capacitor, the potential difference across the combination is 30.0 V. Calculate the unknown capacitance.

3. [____] Tutorial 6 - Question 71

71. Capacitors $C_1 = 6.00\ \mu F$ and $C_2 = 2.00\ \mu F$ are charged as a parallel combination across a 250-V battery. The capacitors are disconnected from the battery and from each other. They are then connected positive plate to negative plate and negative plate to positive plate. Calculate the resulting charge on each capacitor.

4. [____] Tutorial 7 - Question 3

3. A 6.0-μF capacitor charged to 50 V and a 4.0-μF capacitor charged to 34 V are connected to each other, with the two positive plates connected and the two negative plates connected. What is the total energy stored in the 6.0-μF capacitor at equilibrium?
   a. 6.1 mJ
   b. 5.7 mJ
   c. 6.6 mJ
   d. 7.0 mJ
   e. 3.8 mJ